21-11-2017 (E)

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Exam
Nov – Dec 2017

Max. Marks: 100 | **Duration: 3 Hrs**
Class: ~~SY~~ TY | Semester: ~~IV~~ V
Name of the Course: Theoretical Computer Science
Branch: COMP
Course Code: UCEC503

Instructions:
(1) All Questions are Compulsory
(2) Draw neat diagrams
(3) Assume suitable data if necessary

Question No.		Max. Marks
Q 1 (a)	What is finite automata? Give the finite automata M accepting (a,b)*(baaa).	5
Q 1 (b)	Explain Chomsky Hierarchy.	5
Q1 (c)	State and Prove Pumping Lemma for regular languages.	5
Q1(d)	Differentiate between deterministic and non-deterministic PDA.	5
Q2(a)	Construct a finite state machine for binary adder.	10
Q2 (b)	What is TM ? Give the power of TM over FSM. Explain undecidebility and incompleteness in Turing machine. **OR** Explain the variants of Turing Machines.	10
Q3 (a)	Explain CNF and GNF. Convert the following grammar to CNF S -> bA | aB A -> bAA | aS | a B -> aBB | bS | b	10
Q3 (b)	Construct a PDA accepting the following language: $L = \{ a^n b^m a^n \mid m,n \geq 1\}$ **OR** Construct a PDA for the language described as follows: The set of all strings over alphabet {a,b} with exactly equal number of a's and b's.	10

Q4 (a)	Write a regular expression for **any TWO** of the following languages over alphabet {0,1}: 1) Set of all the strings such that the no. of 0's is ODD 2) Set of all the strings that do not contain '110' as a substring. 3) Set of all strings containing all possible combinations of 0's and 1's but not having two consecutive 0's.	10
Q4 (b)	Design a Turing machine that recognizes binary palindromes.	10
Q5 (a)	Write the CFG for the following languages: 1) $L=\{a^n b^m a^n \mid n \geq 0, m \geq 1\}$ 2) L={ All strings of a's and b's with at least two a's.	10
Q5 (b)	Write short notes on **any TWO** of the following: 1) Recursive and recursively enumerable Languages 2) Rice's Theorem 3) Post Correspondence Problem 4) Moore and Mealy Machines.	10

23-11-2017(E)

K. J. Somaiya College of Engineering, Mumbai-77
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End Semester Exam
Nov - Dec 2017

Max. Marks: 100 | **Duration: 3 Hrs**
Class: ~~T.Y. Comp~~ TYBTech | Semester: V
Name of the Course: Advanced Database Management System | Branch: Computer
Course Code: UCEC504

Instructions:
(1) All Questions are Compulsory
(2) Draw neat diagrams
(3) Assume suitable data if necessary

Question No.		Max. Marks
Q.1 (a)	List different phases of database design and implementation process. Explain working of each phase by considering a small example.	10
Q.1 (b)	Define UML Diagram. Explain Class Diagram and state chart diagram with examples. Describe use cases that an automated teller machine (ATM) provides to the bank customers with the help of Use case Diagram. OR Explain Database Tuning with respect to indexes, Database Design and Queries.	10
Q.2 (a)	Explain all three types of Single level Indexing. Compare them. Consider a database of Students with following attribute – 1. Roll No 2. Name 3. Department 4. City Give all three types of Single level indexing for the above table. OR Give difference between Single level and Multilevel indexing. Explain the differences in the implementation of multilevel indexing using B tree and B+ tree.	10
Q.2 (b)	Explain Dynamic hashing for Indexing and give overflow handling technique for dynamic hashing with example.	10
Q.3 (a)	Explain Query processing and measures of Query cost. Calculate Query cost for any three types of join algorithms.	10

Q.3 (b)	Explain External merge sort algorithm with example and give cost calculations.	10
Q.4 (a)	How to parallelize Query evaluation in parallel system? Compare different query evaluation with respect to different partitioning techniques.	10
Q.4 (b)	Explain how Concurrency Control and recovery done in distributed databases in detail. OR Write a short note on object relational features of SQL. Explain nested relational model with example and implementation.	10
Q.5 (a)	Explain Structured, unstructured, semi structured, valid and non-valid XML document with example.	10
Q.5 (b)	Give the difference between DTD and XML Schema. Consider one example of DTD and convert it into XML Schema. OR Explain Querying and transformation of XML data with example.	10

Relational data to

23-11-2017 (E)

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Exam
Nov - Dec 2017

Max. Marks: 20
Class: T.Y.
Name of the Course: Advanced Database Management System
Course Code: UCEC504

Duration: 3 Hrs
Semester: V
Branch: Computer

Instructions:
(1) All Questions are Compulsory
(2) Draw neat diagrams
(3) Assume suitable data if necessary

Question No.		Max. Marks
1	Life cycle of database system is also called A. micro life cycle B. macro life cycle C. mini data life cycle D. meta data life cycle	1
2	In phases of macro life cycle or information system life cycle, phase which consists of study of cost benefit and potential areas of application is called A. deployment analysis phase B. validation analysis phase C. feasibility analysis phase D. implementation analysis phase	1
3	In ordered indices the file containing the records is sequentially ordered, a ___________ is an index whose search key also defines the sequential order of the file. A. Clustered index B. Structured index C. Unstructured index D. Nonclustered index	1
4	In a __________ index, an index entry appears for only some of the search-key values. **A.** Dense **B.** Sparse **C.** Straight **D.** Continuous	1

5	In B+ tree the node which points to another node is called **A.** Leaf node **B.** External node **C.** Final node **D.** Internal node	1
6	In a _______________, the system scans each file block and tests all records to see whether they satisfy the selection condition. **A.** Index Search **B.** Linear search **C.** File scan **D.** Access paths	1
7	Index structures are referred to as ____________, since they provide a path through which data can be located and accessed. **A.** Index Search **B.** Linear search **C.** File scan **D.** Access paths	1
8	Which product is returned in a join query have no join condition: A. Equijoins B. Cartesian C. Both D. None of the mentioned	1
9	Strategy in which few high level entity types splitted into low level entity types is classified as A. top down strategy B. bottom up strategy C. inside out strategy D. mixed strategy	1
10	What does XML stand for? A. eXtra Modern Link B. eXtensible Markup Language C. Example Markup Language D. X-Markup Language	1
11	What is the correct syntax of the declaration which defines the XML version? A. <xml version="A.0" /> B. <?xml version="A.0"?> C. <?xml version="A.0" /> D. None of the above	1

12	What does DTD stand for? A. Direct Type Definition B. Document Type Definition C. Do The Dance D. Dynamic Type Definition	1
13	The use of a DTD in XML development is: A. required when validating XML documents B. no longer necessary after the XML editor has been customized C. used to direct conversion using an XSLT processor D. a good guide to populating a templates to be filled in when generating an XML document automatically	1
14	Location transparency allows for which of the following? **A.** Users to treat the data as if it is at one location **B.** Programmers to treat the data as if it is at one location **C.** Managers to treat the data as if it is at one location **D.** All of the above	1
15	A semijoin is which of the following? **A.** Only the joining attributes are sent from one site to another and then all of the rows are returned. **B.** All of the attributes are sent from one site to another and then only the required rows are returned. **C.** Only the joining attributes are sent from one site to another and then only the required rows are returned. **D.** All of the attributes are sent from one site to another and then only the required rows are returned.	1
16	Some of the column of a relation are at different sites in which of the following ? A. Data Replication B. Horizontal Partitioning C. Vertical Partitioning D. Horizontal and Vertical Partitioning	1
17	The syntax of a user query is verified by A. Query optimizer B. DBA C. Parser D. None of the above	1
18	Collection of information stored in a database at a particular moment is : A. View B. Schema C. Instance **D.** None of the above	1

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19	An abstraction concept for building composite object from the component object is called A. Specialization B. Normalization C. Generalization D. Aggregation	**1**
20	One of the following is a valid record based model : A. Object oriented Model B. Relational Model C. Entity Relationship Model D. None of the above	**1**

23-11-2017(E)

K. J. Somaiya College of Engineering, Mumbai-77
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End Semester Exam
Nov - Dec 2017

Max. Marks: 100 | **Duration: 3 Hrs**

Class: ~~T.Y.~~ Comp TYBTech | Semester: V

Name of the Course: Advanced Database Management System | Branch: Computer

Course Code: UCEC504

Instructions:

(1) All Questions are Compulsory
(2) Draw neat diagrams
(3) Assume suitable data if necessary

Question No.		Max. Marks
Q.1 (a)	List different phases of database design and implementation process. Explain working of each phase by considering a small example.	10
Q.1 (b)	Define UML Diagram. Explain Class Diagram and state chart diagram with examples. Describe use cases that an automated teller machine (ATM) provides to the bank customers with the help of Use case Diagram. OR Explain Database Tuning with respect to indexes, Database Design and Queries.	10
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Q.2 (b)	Explain Dynamic hashing for Indexing and give overflow handling technique for dynamic hashing with example.	10
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Relational data to

25.11.2017(E)

K. J. Somaiya College of Engineering, Mumbai-77
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End Semester Exam
Nov-Dec 2017

Max. Marks: 20 | **Duration: -hrs**
Class: TY.BTech | Semester: V
Name of the Course: Software Engineering (SA) | Branch: Computer Engineering
Course Code: **UCEC505**

Instructions:
(1) All Questions are Compulsory
(2) Draw neat diagrams
(3) Assume suitable data if necessary

Question No.		Max. Marks
Q.1	Distinguish between RAD and Waterfall model (5 points).	5M
Q.2	List types of Agile Methodologies (any 5).	5M
Q.3	Which one of the following testing is performed by the user? a) Acceptance Testing b) Unit Testing c) Compatibility Testing d) None	1M
Q.4	Coupling is a measure of: a) Relative Functional Strength b) Interdependence Among Module c) Both of Above d) None of Above	1M
Q.5	"Consider a system where, a heat sensor detects an intrusion and alerts the security company." What kind of a requirement the system is providing? a) Functional b) Non-Functional c) Known Requirement	1M

Q.6	The type of Software Testing in which each module is tested along in an attempt to discover any errors in its code is: a) Integration Testing b) Acceptance Testing c) Mutation Testing d) Unit Testing	1M
Q.7	Functional requirements capture the intended behaviour of the system. a) True b) False	1M
Q.8	Which of the following diagram is time oriented? a) Collaboration b) Sequence c) Activity	1M
Q.9	Which of the following uses empirically derived formulas to predict effort as a function of LOC or FP? a) FP-Based Estimation b) Process-Based Estimation c) COCOMO d) Both FP-Based Estimation and COCOMO	1M
Q.10	RAD stands for a) Relative Application Development b) Rapid Application Development c) Rapid Application Document	1M
Q.11	Which of the following life cycle model can be chosen if the development team has less experience on similar projects? a) Spiral b) Waterfall c) RAD d) Iterative Enhancement Model	1M
Q.12	Spiral Model has high reliability requirements. a) True b) False	1M

25.11.2017(E)

K. J. Somaiya College of Engineering, Mumbai-77
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End Semester Exam
Nov - Dec 2017

Max. Marks: 100 **Duration:3 hrs**
Class: TYBTech **Semester:** V
Name of the Course: Software Engineering
Branch: Computer
Course Code: UCEC505

Instructions:
(1) **All Questions are Compulsory**
(2) **Assume suitable data if necessary**

Question No.		Max. Marks
Q 1 (a)	Explain RAD for software development with example.	10
	OR	
Q 1 (a)	Explain Agile process with its advantages. Explain any one agile process model	10
Q 1 (b)	Explain the concept of coupling and cohesion. How are they useful in arriving at good software design? Explain with example	10
Q 2 (a)	What do you mean by the requirements? Explain Functional and nonfunctional requirements with suitable example	10
Q 2 (b)	Define metric. Explain function oriented metrics for size estimation of the project	10
	OR	
Q 2 (b)	Explain COCOMO Model in relation to function Point Analysis	10
Q 3 (a)	Explain the mapping of different types of association and generalization relationships to code	10
Q 3 (b)	Explain Version control and Change Control in Software Configuration Management	10
Q 4 (a)	What is the significance of user interface in system development? Draw graphical user interface for online mobile billing	10
Q 4 (b)	Explain the elements used in sequence diagram. Draw sequence diagram for online course registration in our college which checks course availability and student's eligibility before registration	10
	OR	
Q 4 (b)	Assume that Examination software is deployed on client server architecture. Explain various components and its deployment using diagrams	10
Q 5 (a)	Explain the objectives and guidelines for testing. Discuss different performance testing	10
	OR	
Q 5 (a)	Explain Integration and regression testing.	10
Q 5 (b)	Explain software maintenance. Describe different types of software maintenance.	10

16.11.17 (E).

K. J. Somaiya College of Engineering, Mumbai-77

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End Semester Exam

Nov – Dec 2017

Max. Marks:100 | **Duration: 3Hrs.**

Class: TY B.Tech | Semester: V

Name of the Course: Operating System | Branch: Computer

Course Code: UCECE501

Instructions:

(1) All Questions are Compulsory
(2) Draw neat diagrams
(3) Assume suitable data if necessary

Question No.		Max. Marks
Q.1	What are the two models of inter-process communication? What are the strength and weaknesses of the two approaches? **OR** What are the five major activities of an Operating system with regard to file management?	10
Q.2(a)	Assume that the following jobs have to execute with a single processor system, with the jobs arriving in the order listed below where a small integer means a higher priority. <table><tr><td>Process ID</td><td>Service Time</td><td>Priority</td></tr><tr><td>P1</td><td>60</td><td>2</td></tr><tr><td>P2</td><td>20</td><td>1</td></tr><tr><td>P3</td><td>10</td><td>4</td></tr><tr><td>P4</td><td>20</td><td>5</td></tr><tr><td>P5</td><td>50</td><td>3</td></tr></table>Calculate the following for RR (Time Quantum q=10ms), SRTN, Multilevel feedback Queue(number of Queue : 2)→ Time Quantum for Queue-1 is 10 ms Queue-2 is 20 ms a. Create a Gantt chart to illustrate the execution of these processes. b. What is the waiting time and turnaround time for all processes? c. What is the average turnaround time and waiting time.	10
Q.2(b)	What are the advantages and disadvantages of using ULT's instead of KLT's.	5

Q.3(a)	Two processes P1 and P2 have been designed so that P2 writes a stream of bytes produced by P1. Write a skeleton of procedures executed by P1 and P2 to illustrate how they synchronize with one another using semaphore. **OR** An Inventory manager issues following instructions to the store manager in regard to a particular item. Do not purchase the item if the no of items existing in the store exceeds N and hold requisition until the number of items existing in the store is large enough to permit the issue of the item. Using a particular item, implement these instructions with the help of a monitor.	10 10
Q.3(b)	State and explain different mechanisms that can be used to recover a system from deadlock once it is detected. **OR** Explain why spinlocks are not appropriate for single processor system yet are often used in multiprocessor system.	5 5
Q.3(c)	A system is composed of 4 processes [p1,p2,p3,p4] and 3 types of serially reusable resources [R1,R2,R3]. The number of total existing resources are C=[3,2,2]. 1. Process P1 holds 1 unit of R1 and request 1 unit of R2 2. Process P2 holds 2 units of R2 and requests 1 unit each of R1 and R3 3. Process P3 holds 1 unit of R1 and requests 1 unit of R2 4. Process P4 holds 2 units of R3 and requests 1 unit of R1. Show the reusable resource graph to represent this system state. Can the above required resource request granted in the current state If granted, give the order of process execution. If not granted, which processes are in deadlock state?	10
Q.4(a)	Disk track request queue is : 27, 129, 110,186, 147, 41, 10, 64, 120 Assume that the disk head is initially positioned over track 100 and total tracks on disk are 200. Represent the disk head movement graphically and calculate the number of tracks traversed and average seek length in SCAN, LOOK, C-SCAN a. If disk head is moving in the direction of decreasing track numbers. b. If disk head is moving in the direction of increasing track numbers.	10
Q.4(b)	Explain different file organization techniques with the help of diagram. **OR** Describe different file allocation strategies with the help of diagram.	10

Q.4 (c)	Explain different types of I/O buffering techniques with the help of diagrams. **OR** Draw and explain model of I/O organization for Local peripheral devices, communication port and file system.	10
Q.5(a)	Draw and explain 9-state process state transition model used in UNIX.	10
Q.5(b)	Explain the structure of Modern UNIX file system with the help of Diagram.	10

18/11/2017 (E)

K. J. Somaiya College of Engineering, Mumbai-77
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End Semester Examinations
Nov - Dec 2017

Max. Marks: 100
Class: TE
Name of the Course: **Data Networks**
Course Code: **UCE502**

Duration: 3hrs
Semester: V
Branch: Computer

Instructions:
(1) **All Questions are Compulsory**
(2) **Draw neat diagrams**
(3) **Assume suitable data if necessary**

Question No.		Marks
Q 1 (a)	Explain ICMP with packet format. OR Explain ARP with packet format.	10
Q 1 (b)	Explain Mobile IP with header format of IP-N-IP encapsulation	10
Q2 (a)	Explain in brief different issues in IPv4 and IPv6 addressing. OR Explain DHCP in details.	10
Q2 (b)	Explain Hidden and exposed terminal problem in wireless mobile network with the help of neat diagram. How it is solved?	10
Q3 (a)	Compare RIP and OSPF used for intra-domain routing in Autonomous system.	10
Q3 (b)	Explain behavior of DSDV protocol when link is failed after finding shortest path in MANET.	10
Q4 (a)	Explain different ways to improve Quality of Service for a network.	10
Q4 (b)	Explain the role of transport layer in Mobile communication. OR Explain three way handshaking process in TCP.	10
Q5	Explain in detail the following (any two) A]Telnet B]DNS C]WAP D]SMTP	20